

Introduction to Computer Applications (MIS 103)

Lecture 1: Computational Thinking

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What is Computational Thinking ?

- Computers being used in almost every field of life today
- Some pre-requisites for controlling (increasing efficiency of) computers:
 - Understanding of “ how it works?”
 - Making human-computer interaction as efficient as possible

“This process of efficient and effective computer use is known as computer like thinking or **Computational Thinking** [1]

Varying Definitions in Literature

Definition	Source
...the thought processes involved in formulating problems and their solutions so that the solutions are represented in a form that can be effectively carried out by an information-processing agent.	(Cuny, Snyder, Wing, 2010 akt. Wing, 2011, p.20)
Computational thinking is the thought processes used to formulate a problem and express its solution or solutions in terms a computer can apply effectively.	Wing (2014)
The mental process for abstraction of problems and the creation of automatable solutions.	Yadav et al. (2014)
Computational thinking is the process of recognising aspects of computation in the world that surrounds us, and applying tools and techniques from Computer Science to understand and reason about both natural and artificial systems and processes.	Furber (2012)

Varying Definitions in Literature (cont'd)

Definition	Source
Computational thinking has a long history within computer science. Known in the 1950s and 1960s as “algorithmic thinking,” it means a mental orientation to formulating problems as conversions of some input to an output and looking for algorithms to perform the conversions. Today the term has been expanded to include thinking with many levels of abstractions, use of mathematics to develop algorithms, and examining how well a solution scales across different sizes of problems.	Denning (2009)
...[Computational Thinking] is to teach them how to think like an economist, a physicist, an artist, and to understand how to use computation to solve their problems, to create, and to discover new questions that can fruitfully be explored.	Hemmendinger (2010)

What is common? “ Computational Thinking”

- A thought process related to **structured problem solving**
- It involves :
 - Problem understanding and formulation
 - Formulating / designing the solution [in terms a computer can process and solve]
 - Also called “ Algorithmic Thinking” in 1950s and 1960s [**Denning, 2009**]
 - Involves converting inputs to outputs after some processing through algorithms
 - Uses tool and techniques from computer science field
 - Considerations:
 - Many levels of abstractions
 - Use of mathematics to develop algorithms
 - How well a solution scales across different sizes of the problems

Components of Computational Thinking

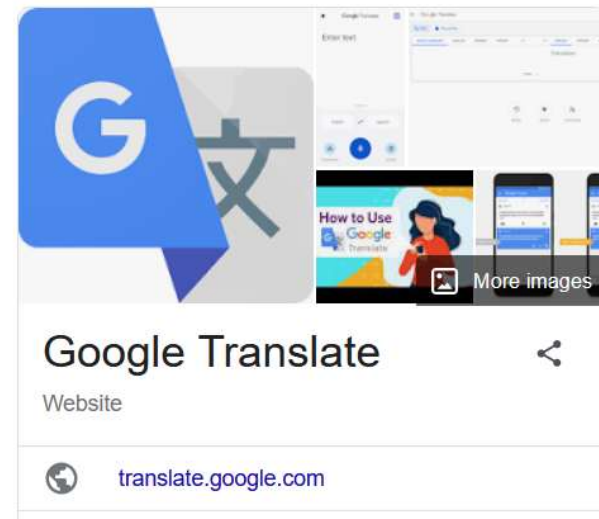
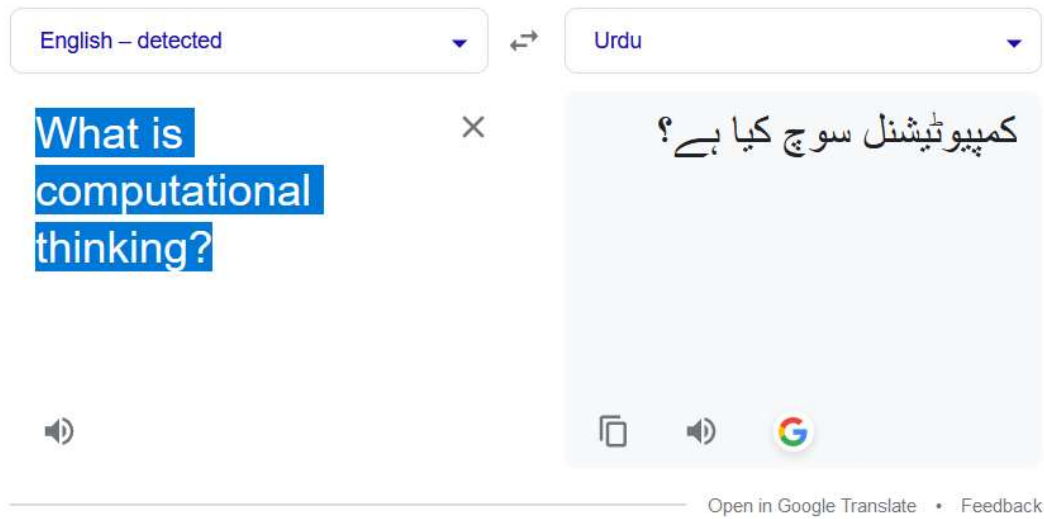
Computational thinking abilities are essentially the set of skills needed to convert complex, messy, partially defined, real-world problems into a form that a mindless computer can tackle without further assistance from a human (BCS, 2014, p.3).

Components	Source
Abstraction, Algorithms, Automation, Problem Decomposition, Parallelization, Simulation	Barr & Stephenson (2011)
Abstraction, Automation, Analysis	Lee et al. (2011)
Abstraction, Algorithmic Thinking, Decomposition, Evaluation, Generalization	Selby & Woollard (2013)
Abstraction, Algorithms, Decomposition, Debugging, Generalization	Angeli et al. (2016)
Abstraction, Algorithms, Automation, Problem Decomposition, Generalization	Wing (2006, 2008, 2011)

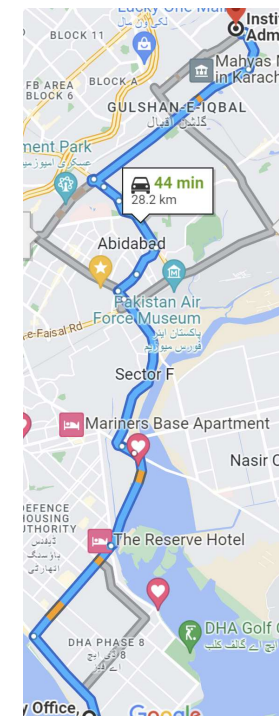
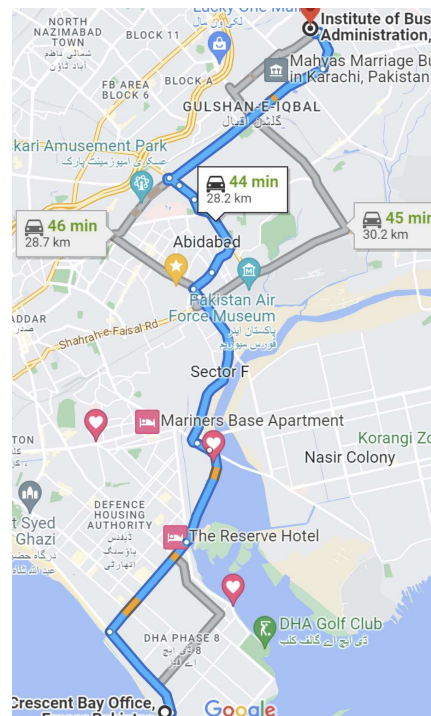
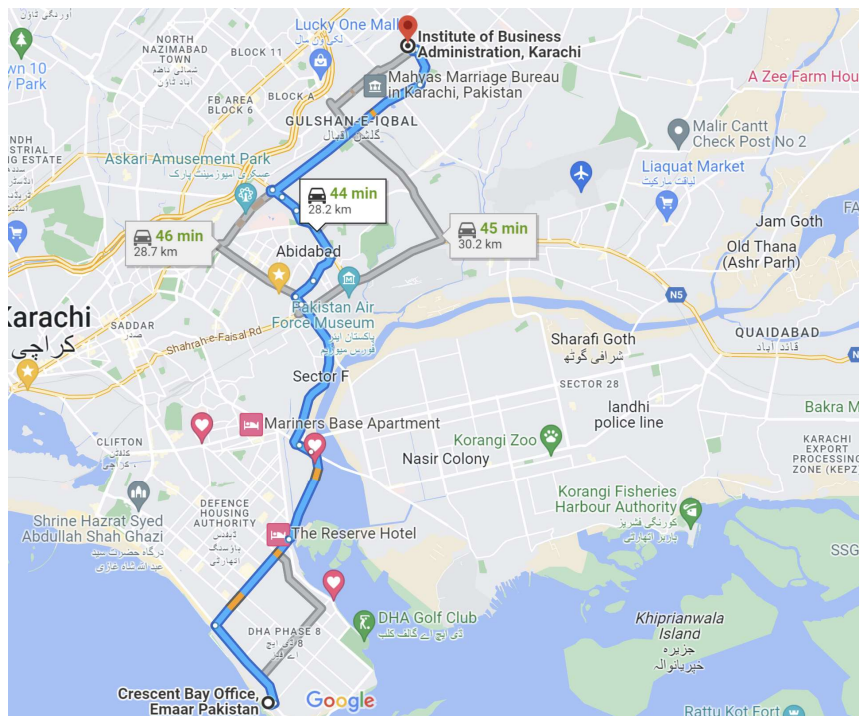
1. Abstraction

- Focusing on the necessary details only while ignoring the unnecessary ones
- Examples
 - Your father asks you for a glass of water
 - Where the focus is?
 - What is abstracted?
 - Your instructor ask you to draw five balls and five stars and separate them with a decision boundary
 - Where the focus is?
 - What is abstracted ?
 - You can send an email by pressing **send** button.
 - Where the focus is?
 - What is abstracted?

Abstraction Examples (Google Translate)



Abstraction Examples (Google Map)



What is hidden?

Google Translate?

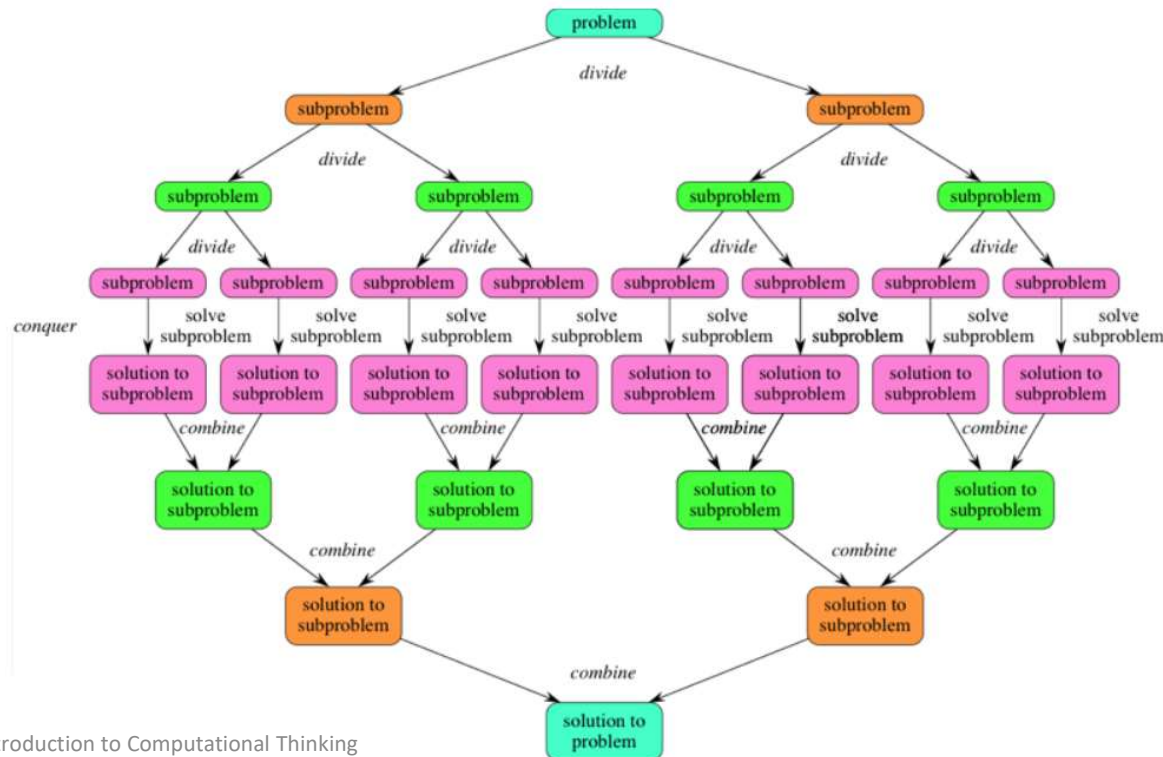
- ?
- ?
- ?
- ?

Google Maps?

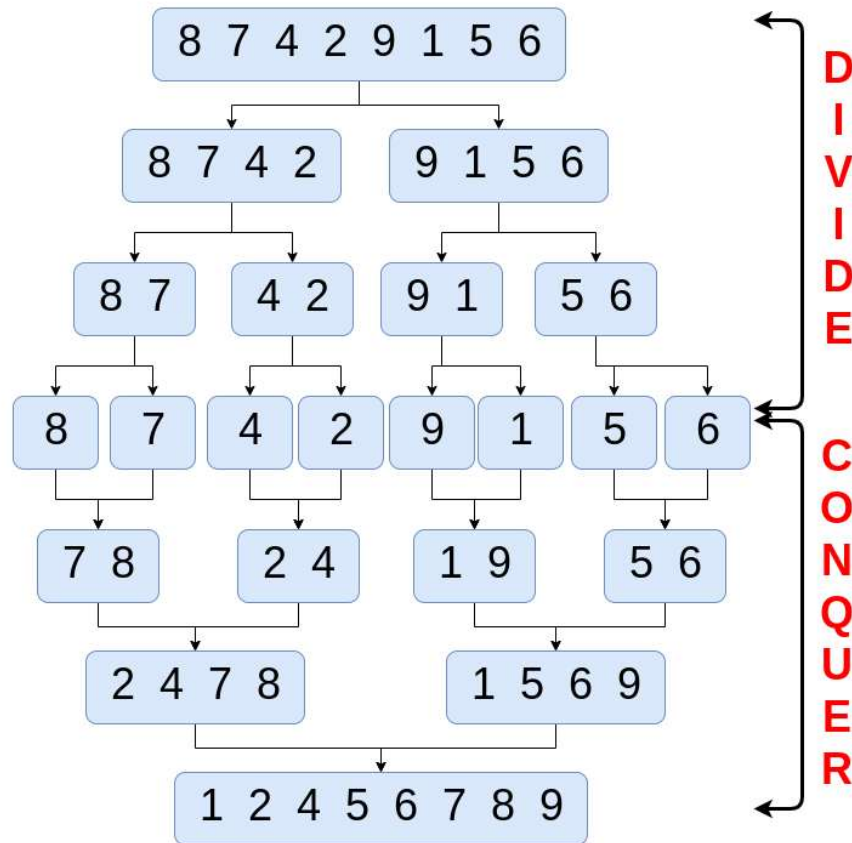
- ?
- ?
- ?
- ?

2. Problem Decomposition

- **Problem Decomposition** is a method for taking a problem and breaking it into smaller and more understandable constituents.
- This method is also known as “**Divide and Conquer**”

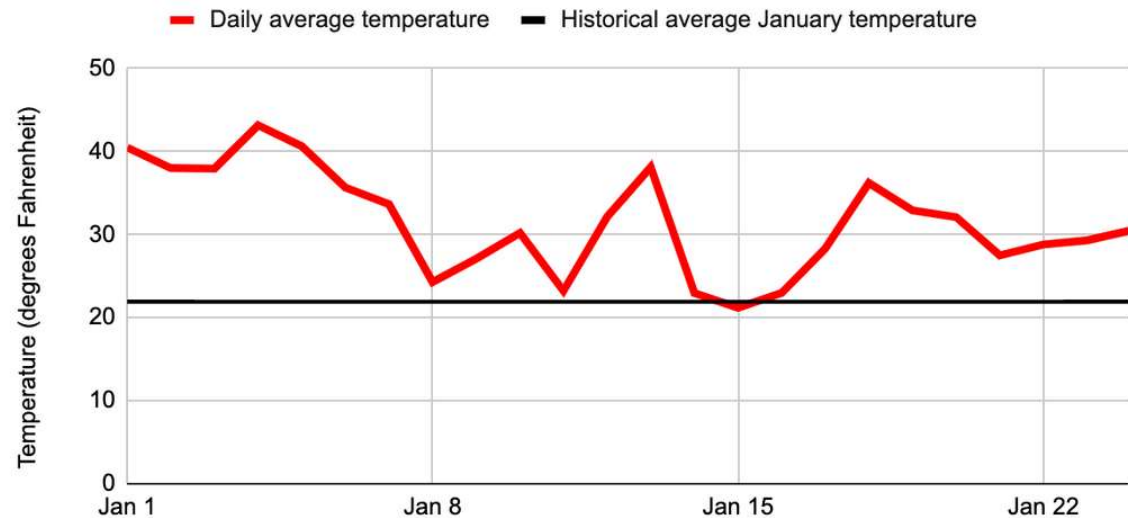


Problem Decomposition-Divide and Conquer (Merge Sort)



Problem Decomposition (Input – Processing – Output)

- Writing a computer program to compute the average daily temperatures of the previous 3 weeks and display it graphically.
- Decompose the problem
 - Input
 - Processing
 - Output



Problem Decomposition -Work Breakdown Structure

- A **Work Breakdown Structure (WBS)** is a project management tool that helps break down complex projects into smaller, more manageable components.
- It essentially acts like a roadmap for your project, showing the overall scope and identifying all the tasks needed to complete it successfully.

Here are some examples of how a WBS can be used in different projects:

- **Developing a new software application:** The WBS might include sub-deliverables like design documents, code modules, and user manuals.
- **Building a house:** The WBS might include sub-deliverables like foundation, framing, plumbing, and electrical work.
- **Marketing a new product:** The WBS might include sub-deliverables like market research, product design, advertising campaign, and sales launch.

WBS

Work Breakdown Structure Template

Project Name **Project Start Date**

Project Owner: Client:
 Address: Address:
 Email: Email:
 Phone: Phone:

Level	WBS	Description	Assigned To	Start Date	End Date	Notes
1	1	Stage 1	Sr.Manager			
2	1.1	Level 2 Description	Manager			
2	1.2	Level 2 Description	Manager			
3	1.2.1	Level 3 Description	Asst.Manager			
4	1.2.1.1	Level 4 Description	Analyst			
4	1.2.1.2	Level 4 Description	Analyst			
1	2	Stage 2	Project Engineer			
2	2.1	Level 2 Description	Senior Engineer			
3	2.1.1	Level 3 Description	Junior Engineer			
3	2.1.2	Level 3 Description	Junior Engineer			
4	2.1.2.1	Level 4 Description	Trainee			

Decomposition – Example 1

Write a Haiku

To be concise, poetic and to the point in 3 lines and 17 syllables. There is a beauty in constraints.

<http://writeahaiku.com>

Haiku rules

- Three lines.
- Five syllables in the first and third lines.
- Seven syllables in the second line.
- Lines don't rhyme.
- Includes a kireji or cutting word. (but , so , yet , then, or etc.)
- Includes a kigo, a seasonal reference. (cool , warmth, flowering, fallen leaves etc)

Haiku - Rules

rain pitter patters
so very quiet and sad
so warm and calming

beautiful flowers
take care of them and they grow
neglect them, they die

Decomposition : Examples 2 - Write a speech

Tell them what you are going to tell them.

Tell them.

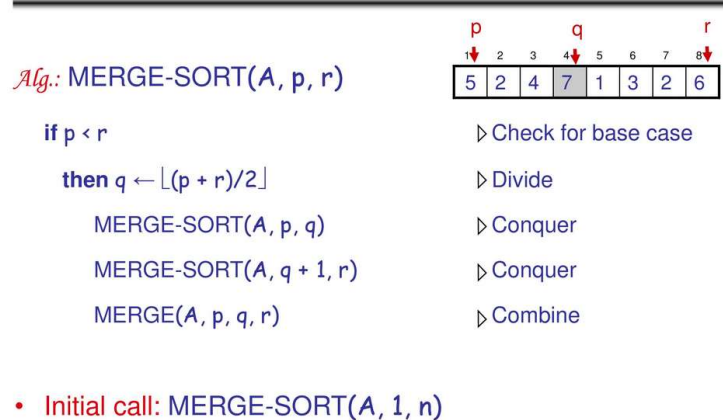
Tell them what you have told them.

3. Algorithm Design

- An algorithm is a step-by-step procedure for solving a problem.
- Guidelines
 - **Correctness:** The algorithm must be correct, meaning that it must produce the correct output for all possible inputs.
 - **Efficiency:** The algorithm should be efficient, meaning that it should solve the problem in a reasonable amount of time and space.
 - **Generality:** The algorithm should be general, meaning that it should be able to solve a wide variety of problems.
 - **Simplicity:** The algorithm should be simple, meaning that it should be easy to understand and implement.
 - **Robustness:** The algorithm should be robust, meaning that it should be able to handle unexpected inputs and errors.

Algorithm Design

Merge Sort



- Assume for instance that the gross pay of an employee is to be calculated and then 10 percent of the gross is to remove as tax. The yield being the net pay. Write down the algorithm for this problem.

Sol.:

1. Begin
2. input name, hours_worked, and wage par hour
3. calculate gross_pay = hours_worked * wage par hour
4. calculate tax = (10/100) * gross_pay
5. calculate net_pay = gross_pay - tax
6. print name, net_pay
7. End

Searching

- How will you search an element in an unordered list
 - Searching an email address in a list of 50 with out any index
- How will you search an element in an ordered or sorted list
 - Searching a name in telephone directory and then the phone number
 - Searching a word in a dictionary
- Considerations:
 - Efficiency in terms of execution time
 - Complexity of the algorithm

Linear Search

The procedures for implementing linear search are as follows:

Step 1: First, read the search element (Target element) in the array.

Step 2: In the second step compare the search element with the first element in the array.

Step 3: If both are matched, display "Target element is found" and terminate the Linear Search function.

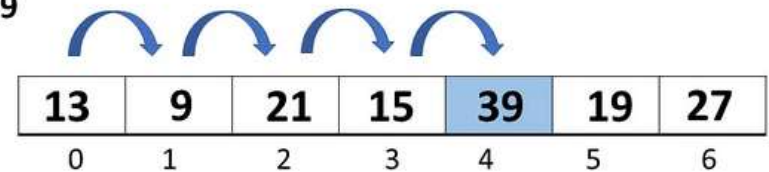
Step 4: If both are not matched, compare the search element with the next element in the array.

Step 5: In this step, repeat steps 3 and 4 until the search (Target) element is compared with the last element of the array.

Step 6 - If the last element in the list does not match, the Linear Search Function will be terminated, and the message "Element is not found" will be displayed.

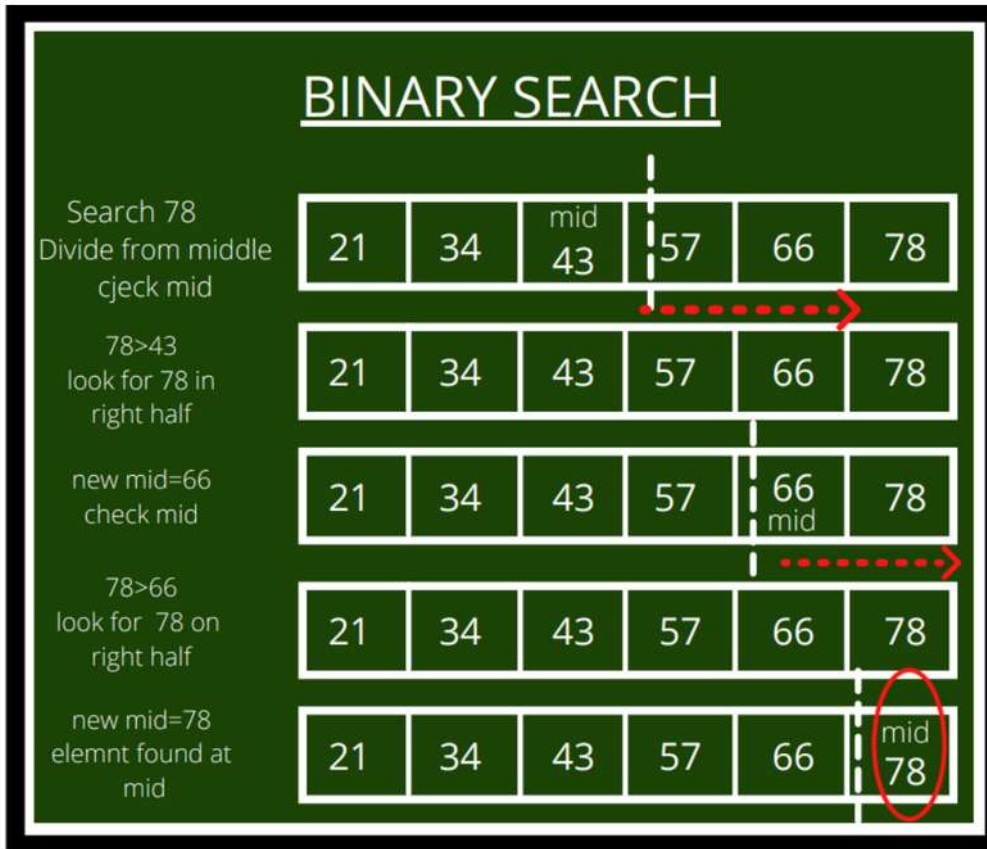
Searched Element

39



13	9	21	15	39	19	27
0	1	2	3	4	5	6

Binary Search



Binary Search – Algorithm

BINARY SEARCH			Array Divide and Conquer
Best	Average	Worst	
$O(1)$	$O(\log n)$	$O(\log n)$	

search (A, t)
1. $low = 0$
2. $high = n - 1$
3. **while** ($low \leq high$) **do**
4. $ix = (low + high) / 2$
5. **if** ($t = A[ix]$) **then**
6. **return true**
7. **else if** ($t < A[ix]$) **then**
8. $high = ix - 1$
9. **else** $low = ix + 1$
10. **return false**
end

search (A, 11)

first pass

<i>low</i>				<i>ix</i>			<i>high</i>
1	4	8	9	11	15	17	

second pass

				<i>low</i>	<i>ix</i>	<i>high</i>	
1	4	8	9	11	15	17	

third pass

				<i>low</i>	<i>ix</i>	<i>high</i>	
1	4	8	9	11	15	17	

explored elements

4. Pattern Recognition

- Ability to **identify and understand similarities and relationships within data or concepts.**
 - Look for repeating sequences, shapes, relationships, or trends in data.
 - Analyze common features of different cases within a problem.
 - Compare and contrast different parts of the problem to recognize similarities and differences.

Pattern Recognition - Examples

1. Identifying seasonal trends in sales data:

- **Data:** You have a dataset of sales figures for a clothing store for the past year.
- **Patterns to look for:**
 - **Seasonal:** Are there any sales spikes or dips that coincide with specific seasons, like increased winter jacket sales or summer swimsuits?
 - **Holidays:** Do certain holidays lead to predictable sales rises, like Mother's Day for dresses or Black Friday for electronics?
 - **Weekday/weekend differences:** Are there consistent differences in sales between weekdays and weekends?

Pattern Recognition – Examples

2. Detecting fraudulent claims in insurance data:

- **Data:** You have a dataset of insurance claims filed by customers.
- **Patterns to look for:**
 - **Unusual claim frequency:** Does a particular customer submit claims unusually often compared to others?
 - **Suspicious claim amounts:** Are there claims with excessively high amounts for the type of damage reported?
 - **Claims with similar characteristics:** Are there groups of claims with similar locations, types of damage, or timing, potentially indicating coordinated fraud?

Pattern Recognition

- It is the ability to identify patterns in historical data and use those patterns to solve future problems.
- Pattern recognition can be used to solve a wide variety of problems, including:
 - Identifying objects in images
 - Classifying text
 - Predicting future events
 - Detecting fraud
 - Making medical diagnoses

Pattern Recognition – Traffic Patterns

DHA to IBA

- At 6 am
- At 8 am
- At 6 pm
- At 9 pm
- On a weekend

5. Generalization

- Generalization is the process of adapting formulated solutions or algorithms into different problem states, even if the variables involved are different.
- Example
 - You are required to write a program that sorts a list of numbers
 - Suppose you write a program to sort numbers (integers) in ascending order
 - Limitations:
 - Can't work with float values
 - Can't sort in descending order
 - May have limit in size of the list provided
 - A general solution can be to design an algorithm that can:
 - Work with both integers and floats
 - Gives the user the flexibility to sort in any order
 - Can work with any size of list

What does it take to get things done?

Solving problems

An objective.
An end goal

Constraints,
limitations and
restrictions

A history of solving similar problems?

Abstraction

Decomposition

Algorithm
Design

Pattern
Recognition

Generalization

What does it take to get things done?

Frameworks

- Input. Output. Process
- Smaller sub problems. Divide and conquer.
- Work break down structure. Task lists

Process

- Abstraction
- Pattern recognition
- Decomposition
- Design
- Generalization

Deliverables

- Efficiency
- Environment
- Constraints
- Deployment

Assessments



Lecture 1: Introduction to Computational Thinking



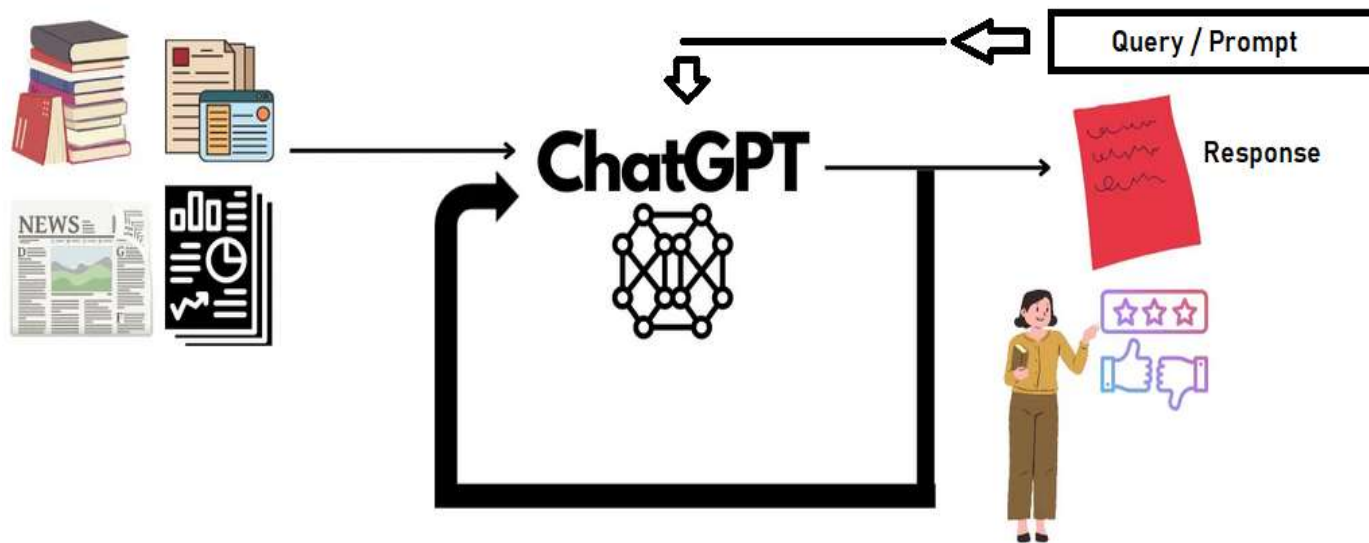
Business Administration
Karachi

Leadership and Ideas for Tomorrow 35

How to get A Grade in a course

- Problem : How to get A grade in a Course
 - Abstraction
 - Decomposition
 - Algorithm Design
 - Pattern Recognition
 - Generalization

How does ChatGPT Work?



ChatGPT : How it works?

- Suppose , someone writes this prompt

• *“The best thing about AI is its ability to”*

ChatGPT : How it works

- ChatGPT is always fundamentally trying to do is to produce a “reasonable continuation” of whatever text it’s got so far.
- By “reasonable” we mean “what one might expect someone to write after seeing what people have written on billions of webpages, etc.”
- Imagine scanning billions of pages of human-written text (say on the web and in digitized books) and finding all instances of this text, it produces a ranked list of words that might follow, together with “probabilities”:

<i>The best thing about AI is its ability to</i>	learn	4.5%
	predict	3.5%
	make	3.2%
	understand	3.1%
	do	2.9%

ChatGPT : How it works

- At each step it gets a list of words with probabilities.
- One might think the next word should be the “highest-ranked”
 - If we always pick the highest-ranked word, we’ll typically get a very “flat” essay, that never seems to “show any creativity”
 - But if sometimes (at random) we pick lower-ranked words, we get a “more interesting” essay.
 - Not using the highest ranked word helps in creating randomness
- Temperature Parameter
 - Decides how often a lower-ranked word will be used
 - Values close to 0 generate more exact response with less randomness
 - Values close to 1 tend to generate something novice

ChatGPT : How it works

```
In[ ]:= model["The best thing about AI is its ability to", {"TopProbabilities", 5}]
```

```
Out[ ]:= { do → 0.0288508, understand → 0.0307805,  
          make → 0.0319072, predict → 0.0349748, learn → 0.0445305 }
```

This takes that result and makes it into an explicit formatted “dataset”:

```
In[ ]:= Dataset[ReverseSort[Association[%]],  
               ItemDisplayFunction → (PercentForm[#, 2] &)]
```

learn	4.5%
predict	3.5%
make	3.2%
understand	3.1%
do	2.9%

ChatGPT: How it works?

Here's what happens if one repeatedly “applies the model”—at each step adding the word that has the top probability (specified in this code as the “decision” from the model):

```
In[ ]:= NestList[StringJoin[#, model[#, "Decision"]]&,  
  "The best thing about AI is its ability to", 7]
```

```
Out[ ]:= {The best thing about AI is its ability to,  
  The best thing about AI is its ability to learn,  
  The best thing about AI is its ability to learn from,  
  The best thing about AI is its ability to learn from experience,  
  The best thing about AI is its ability to learn from experience.,  
  The best thing about AI is its ability to learn from experience. It,  
  The best thing about AI is its ability to learn from experience. It's,  
  The best thing about AI is its ability to learn from experience. It's not}
```

ChatGPT: How it Works?

What happens if one goes on longer? In this (“zero temperature”) case what comes out soon gets rather confused and repetitive:

The best thing about AI is its ability to learn from experience. It's not just a matter of learning from experience, it's learning from the world around you. The AI is a very good example of this. It's a very good example of how to use AI to improve your life. It's a very good example of how to use AI to improve your life. The AI is a very good example of how to use AI to improve your life. It's a very good example of how to use AI to

ChatGPT: How it Works?

But what if instead of always picking the “top” word one sometimes randomly picks “non-top” words (with the “randomness” corresponding to “temperature” 0.8)? Again one can build up text:

```
{ The best thing about AI is its ability to,  
  The best thing about AI is its ability to create,  
  The best thing about AI is its ability to create worlds,  
  The best thing about AI is its ability to create worlds that,  
  The best thing about AI is its ability to create worlds that are,  
  The best thing about AI is its ability to create worlds that are both,  
  The best thing about AI is its ability to create worlds that are both exciting,  
  The best thing about AI is its ability to create worlds that are both exciting, }
```

ChatGPT: How it Works?

And every time one does this, different random choices will be made, and the text will be different—as in these 5 examples:

The best thing about AI is its ability to learn. I've always liked the

The best thing about AI is its ability to really come into your world and just

The best thing about AI is its ability to examine human behavior and the way it

The best thing about AI is its ability to do a great job of teaching us

The best thing about AI is its ability to create real tasks, but you can

ChatGPT: How it Works?

This was done with the [simplest GPT-2 model](#) (from 2019). With the newer and [bigger GPT-3 models](#) the results are better. Here's the top-word (zero temperature) text produced with the same “prompt”, but with the biggest GPT-3 model:

The best thing about AI is its ability to automate processes and make decisions quickly and accurately. AI can be used to automate mundane tasks, such as data entry, and can also be used to make complex decisions, such as predicting customer behavior or analyzing large datasets. AI can also be used to improve customer service, as it can quickly and accurately respond to customer inquiries. AI can also be used to improve the accuracy of medical diagnoses and to automate the process of drug discovery.

And here's a random example at “temperature 0.8”:

The best thing about AI is its ability to learn and develop over time, allowing it to continually improve its performance and be more efficient at tasks. AI can also be used to automate mundane tasks, allowing humans to focus on more important tasks. AI can also be used to make decisions and provide insights that would otherwise be impossible for humans to figure out.

Visual Thinking

Contents

- What is Visual Thinking?
- Visual Thinking Tools
- Mind Maps
- Control Flow Charts
- Process Flow Diagrams
- Swim Lanes

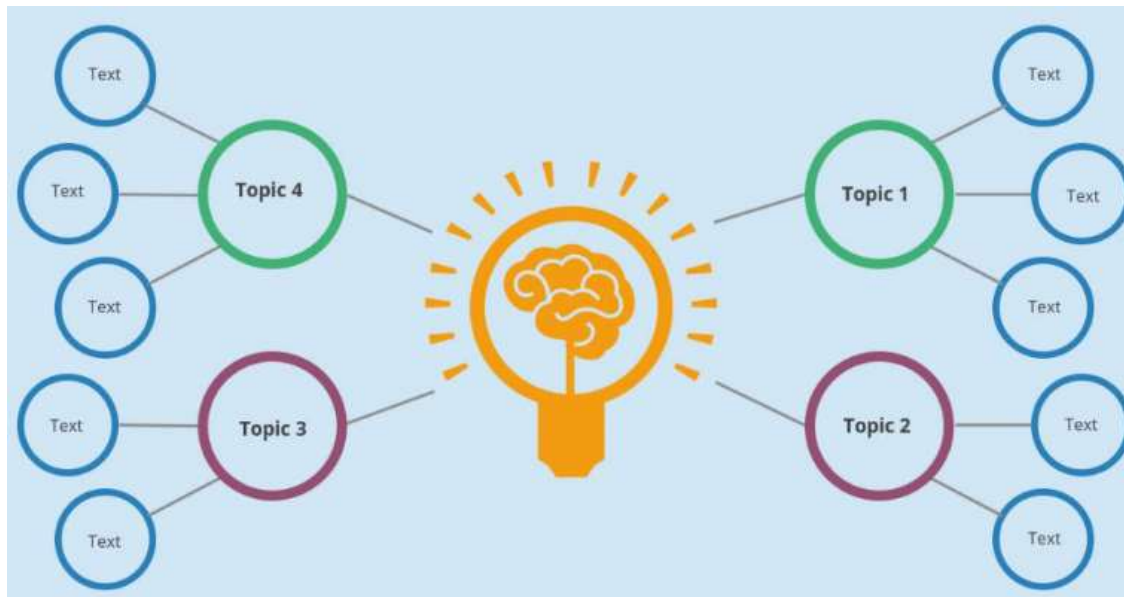
Visual Thinking

- Visual thinking is a thought process that organizes ideas visually and focuses on graphic representation instead of a verbal representation of information or problem.
- Visual thinking helps in:
 - Stimulate problem-solving
 - Break down complex structures of information
 - Create strong associations between pieces of information
 - Explore innovative ideas
 - Organize ideas graphically

Visual Thinking

- The following tools and techniques can be used:
 - Mind Maps
 - Process Flow Diagrams
 - Flow Charts
 - Visual Programming
 - Swim Lanes
 - State Diagrams
 - Transition Tables
 - UML Diagrams

What is Mind Mapping?

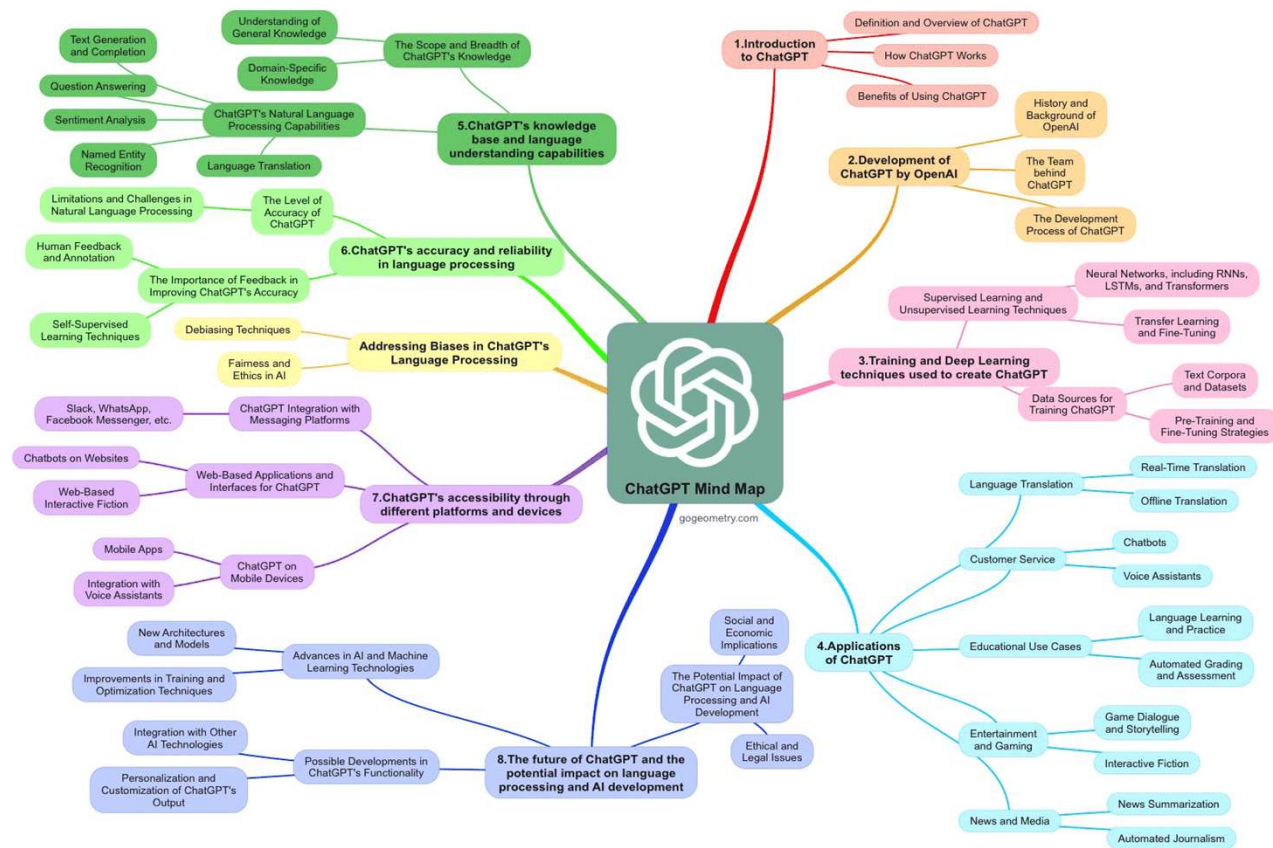


“A visual thought process that assists you to identify a central idea, brainstorm a cloud of related ideas, and creatively think about specific dimensions of the main idea to help you see that you are considering all of its benefits or solution strategies”

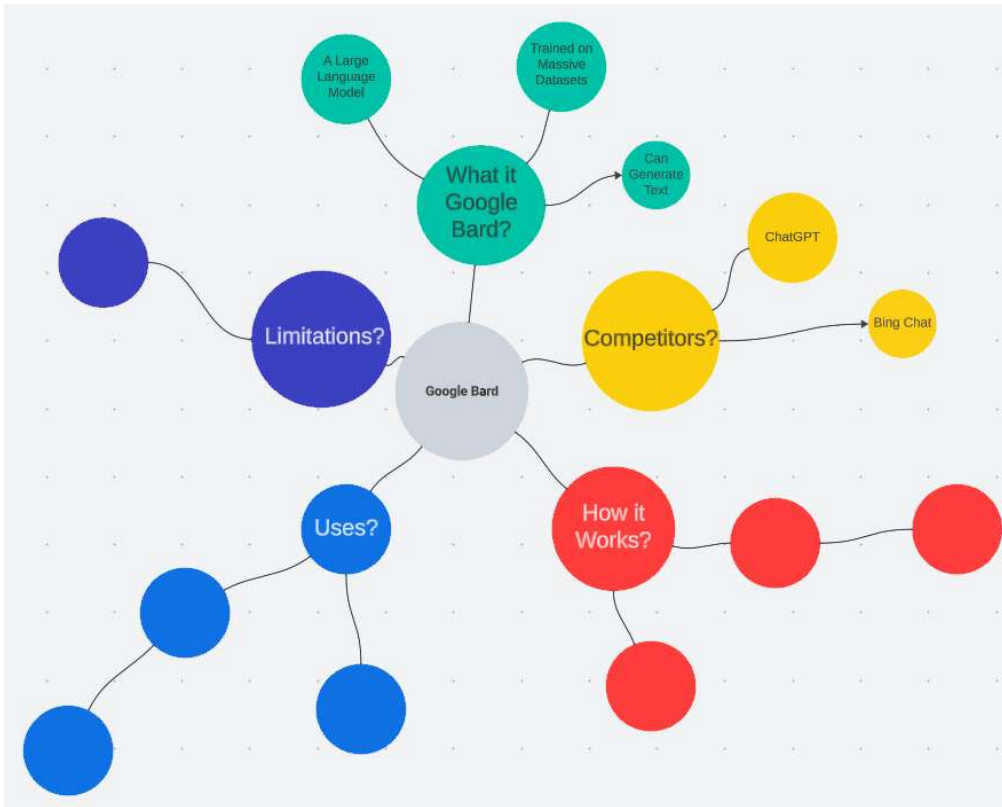
Mind Map – Key Element

- Here are some of the key elements of a mind map:
 - A central image or concept.
 - Branches that radiate from the central image, representing related ideas.
 - Sub-branches that radiate from the branches, representing more specific ideas.
 - Images, words, and colors can be used to represent ideas.
 - Symbols can be used to represent relationships between ideas.

Exploring ChatGPT using a Mind Map



TO DO : Create a mind map for Google Bard?



Control Flow Diagrams

What is a control flow diagram ?

- A control flow diagram (CFD) is a graphical representation that shows the complete flow of execution of a computer program or algorithm.
- It shows:
 - The inputs
 - Processing paths based on different logical conditions
 - The Output
- It contains:
 - Nodes: Nodes represent points in the program where control can be transferred.
 - Arrows: Arrows represent the flow of control between nodes.
 - Decision nodes: Decision nodes represent points in the program where a decision needs to be made.
 - Terminal nodes: Terminal nodes represent the end of the program.

Control Flow Diagram- Basic Symbols

Symbol	Name	Function
	Start/end	An oval represents a start or end point
	Arrows	A line is a connector that shows relationships between the representative shapes
	Input/Output	A parallelogram represents input or output
	Process	A rectangle represents a process
	Decision	A diamond indicates a decision

Linear Search – Flow Chart

Here are the steps of the flowchart:

Step 1) Read the search item, “item.”

Step 2) Initiate $i=0$ and $\text{index}=-1$

Step 3) If $i < N$, go to step 4. Else, go to step 8.

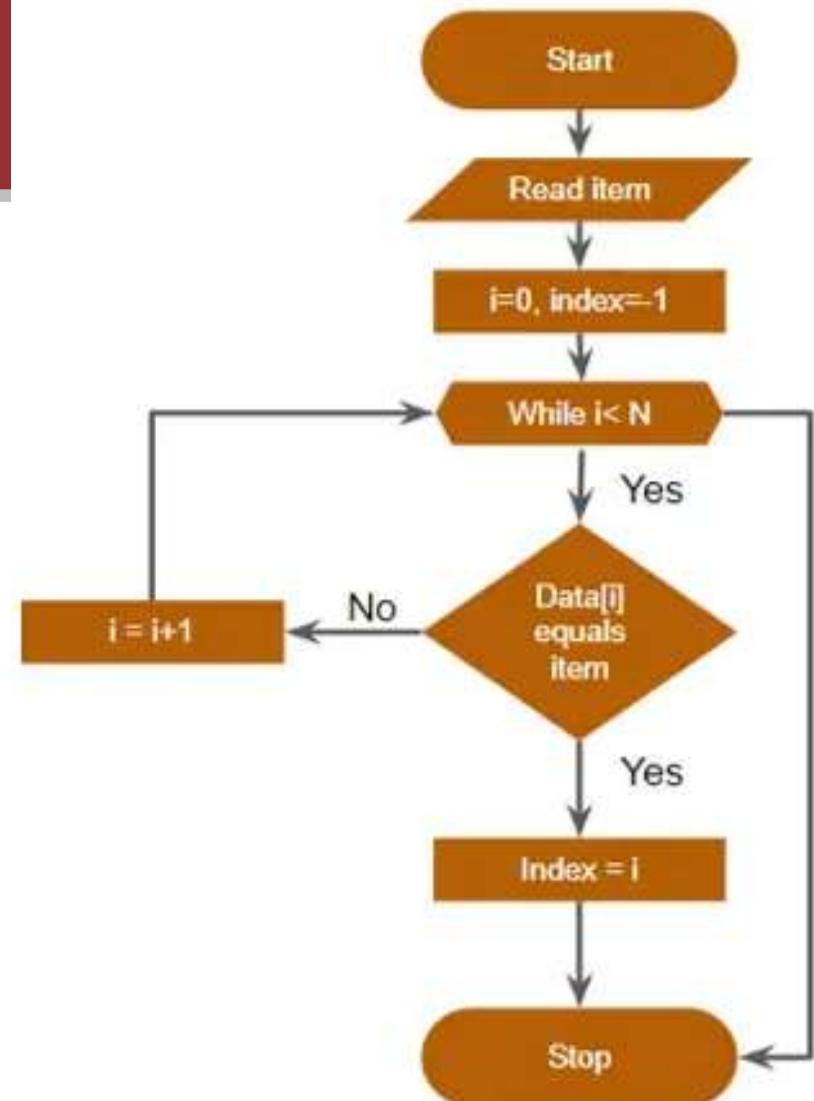
Step 4) If $\text{Data}[i]$ equals to “item,” then go to step 5. Else go to step 6.

Step 5) $\text{Index} = i$ (As the item is found at index no i). Go to step 8.

Step 6) $i = i + 1$.

Step 7) Go to step 3.

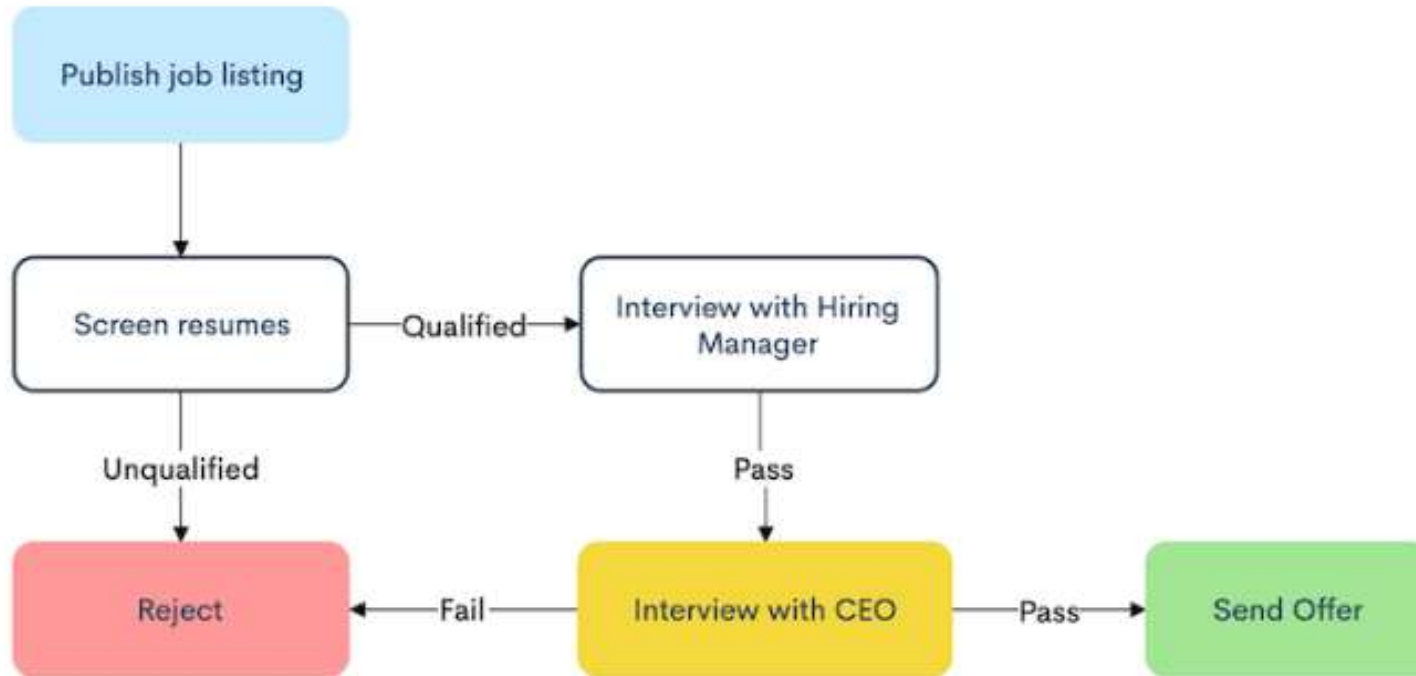
Step 8) Stop.



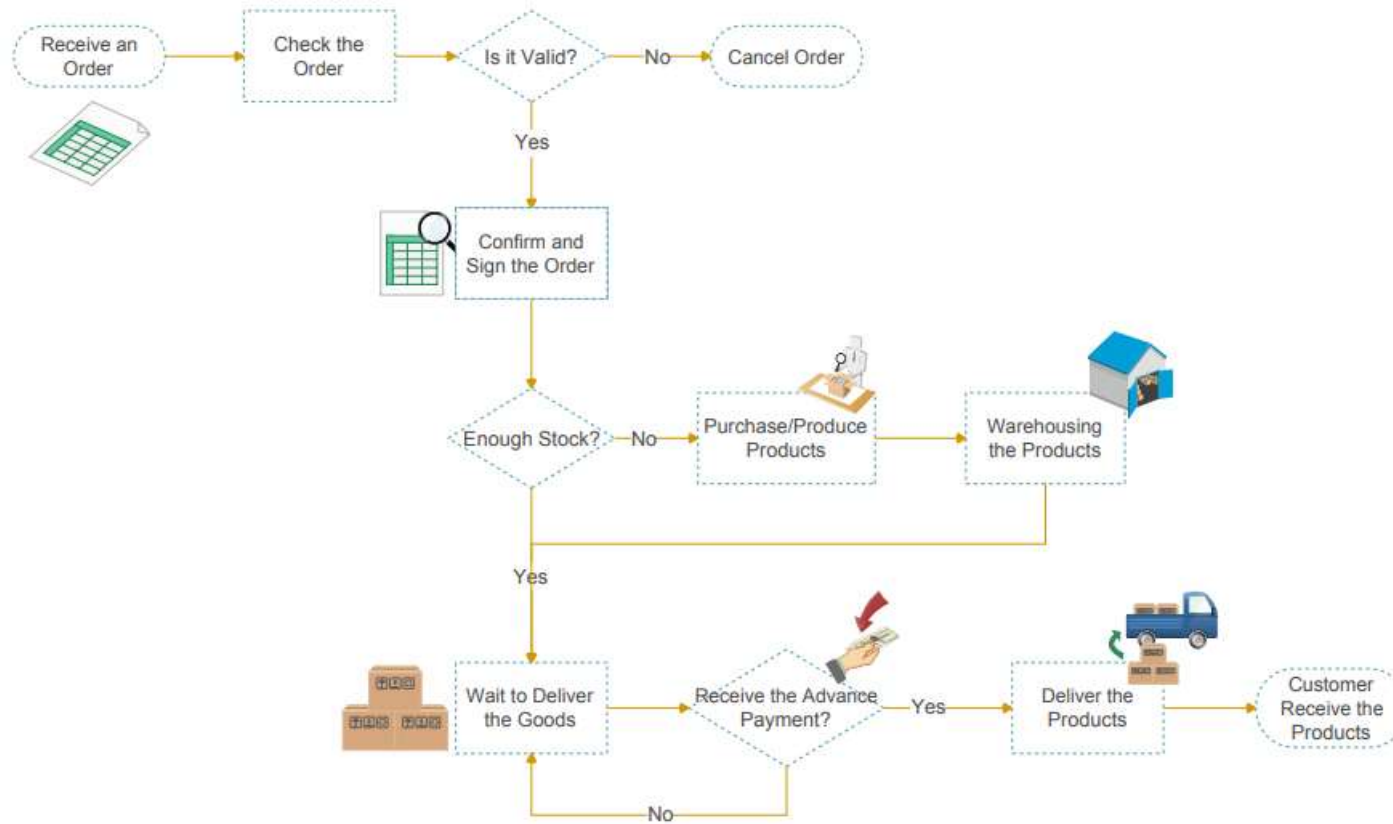
Process Flow Diagrams

- **A process flow diagram is a flowchart that illustrates the sequence of steps needed to complete a business process.**
- **It uses standardized symbols to describe and depict the flow between each step.**
- **Application:** Process flow diagrams are used in project management, business processes, and software development to visualize workflows and identify bottlenecks.

Process Flow Diagram – Hiring Process



Process Flow Diagram: Order Processing



Steps for making a business process flow diagram

1. Determine the main components of the process.
2. Order the activities
3. Choose the correct symbols for each activity
4. Make the connection between the activities
5. Indicate the beginning and end of the process
6. Review your business process diagram

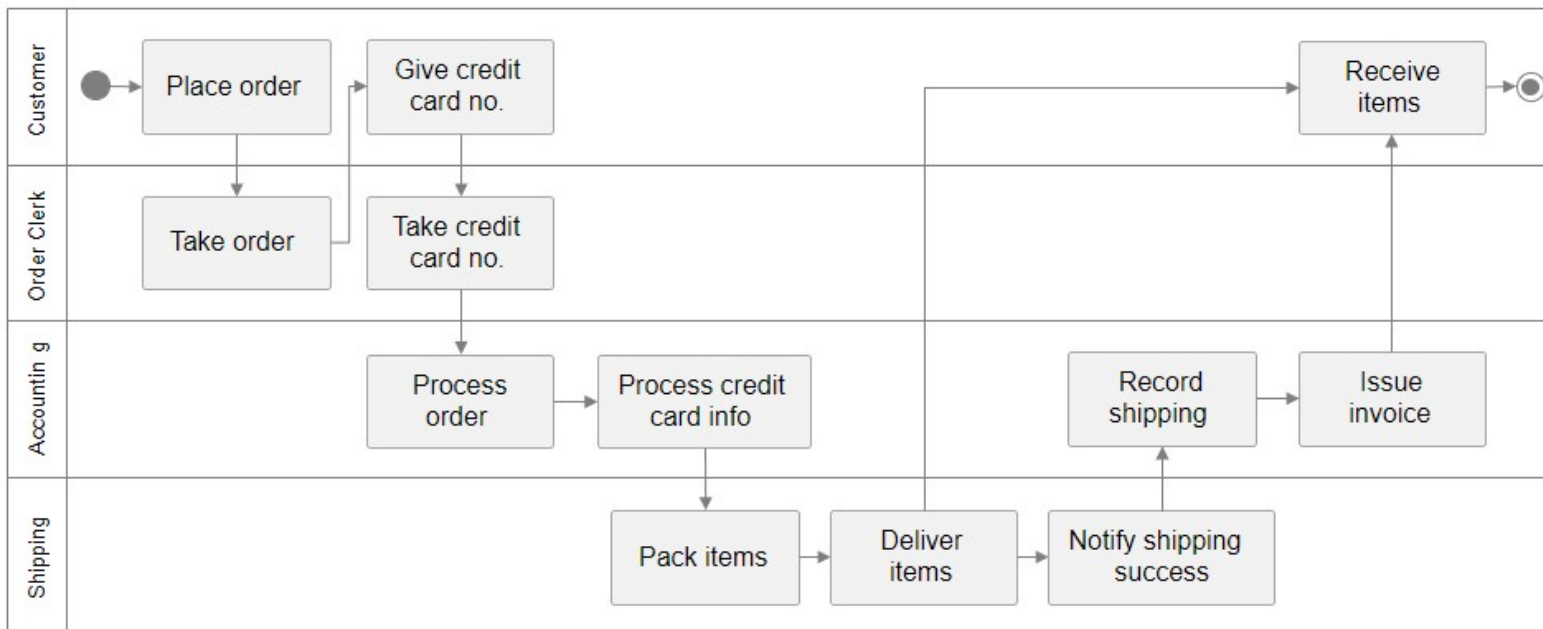
Swim Lanes

Swim Lanes

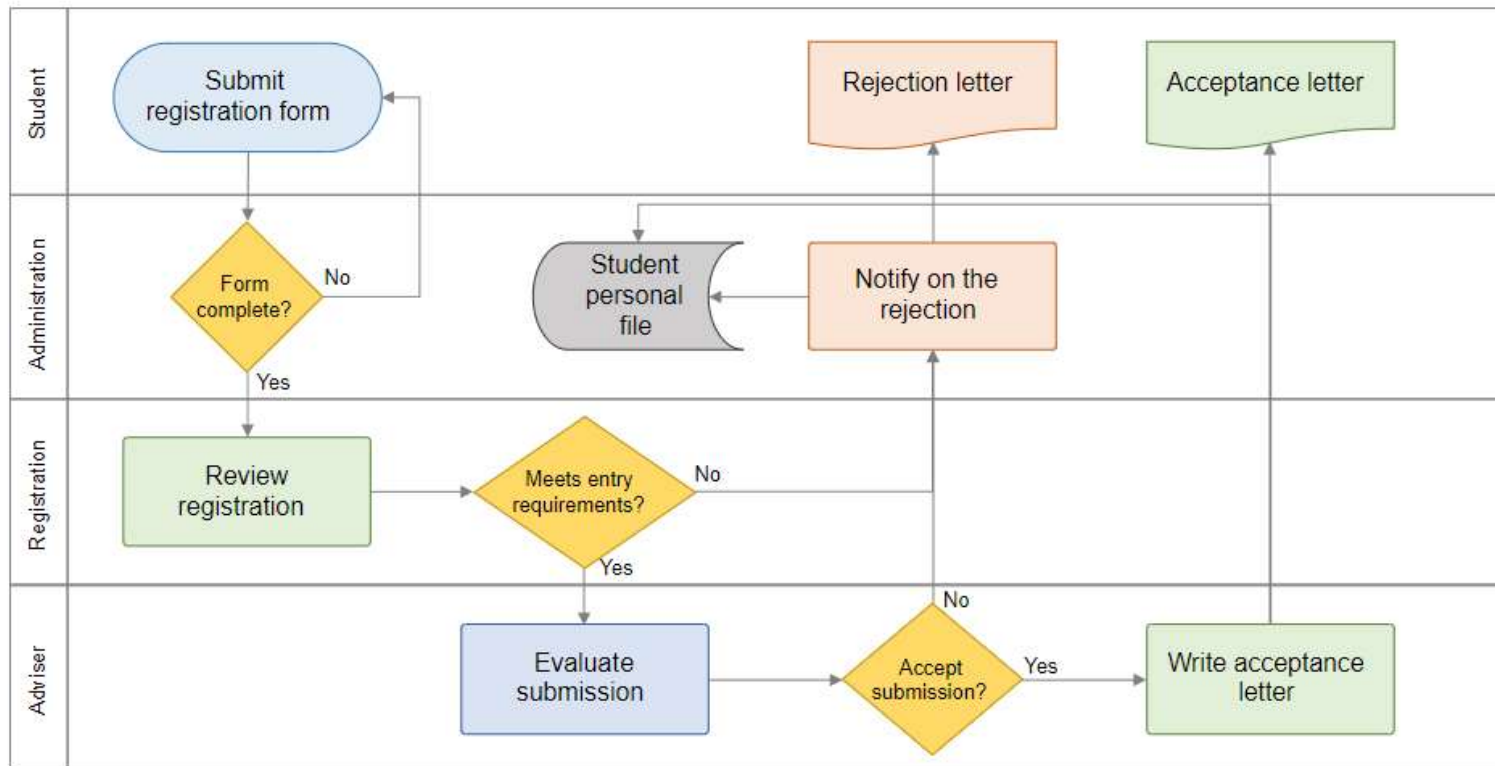
- A swim lane diagram delineates who does what in a process.
- Using the metaphor of lanes in a pool, a swim lane diagram provides clarity and accountability by placing process steps within “swim lanes” of a particular employee, work group or department.

Application: They are valuable in business process analysis and team collaboration scenarios to understand roles and handoffs.

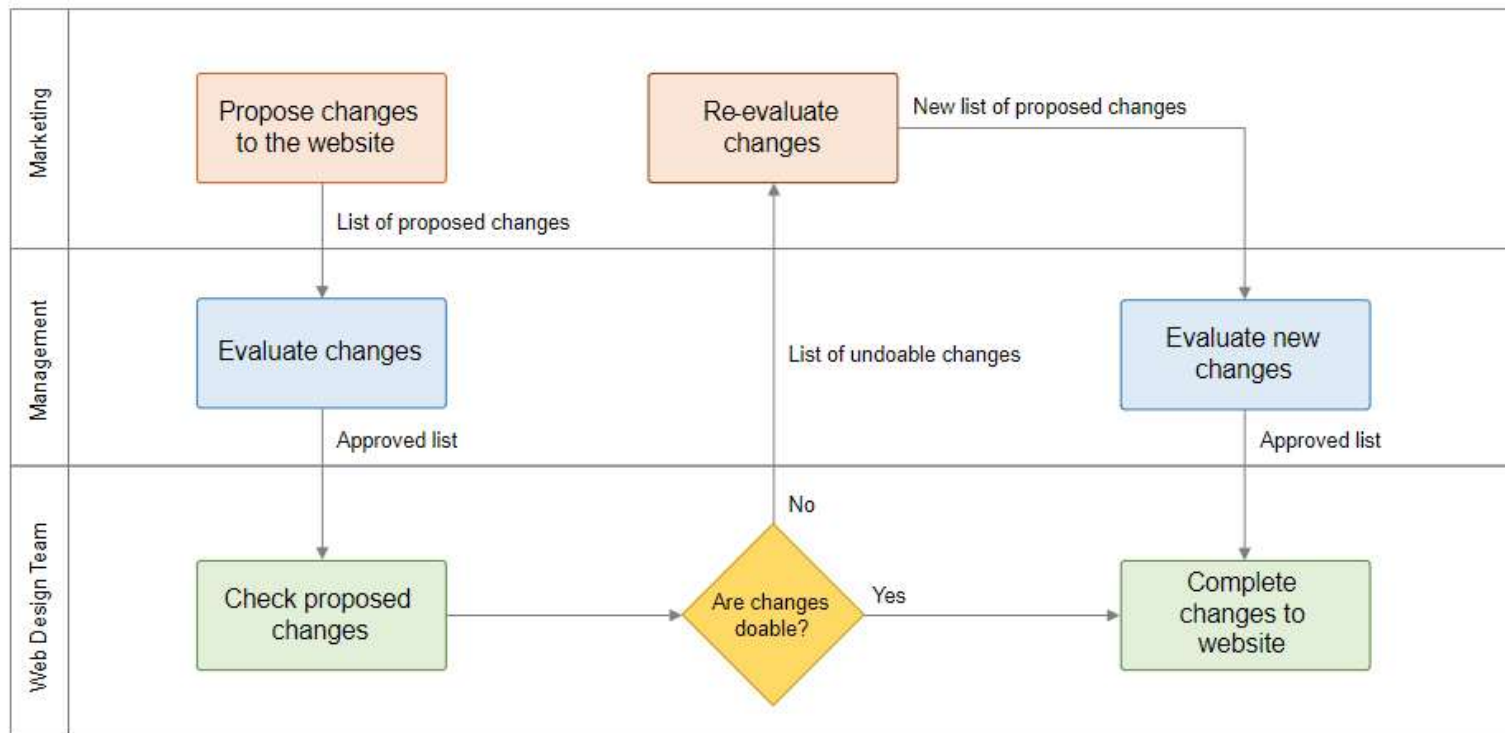
Order Processing



Student Registration



Website Changes



Week 1 - Reading Assignment

- Chapter 1 : Introduction to Computers and Programming
(Book 1 : Starting out with Python)

References

- [1] [An overview of computational thinking.](#)
- [2] [The long quest for computational thinking.](#)
- [3] [What is ChatGPT and How it works?](#)
- [4] <https://www.cybermedian.com/pl/comprehensive-guide-to-flowchart-with-50-examples/>
- [5] <https://www.officetimeline.com/swimlane-diagram>